

A letter stands for a number we do not know yet. Simplify each expression by collecting like terms. Show your working in your workbook.

Method

- 1 Find the like terms (same letter)
- 2 Group them, keeping each sign
- 3 Add or subtract the number parts
- 4 Letters first, then the number
- 5 Check by substituting a value

Common mistakes

- $3x$ means $3 \times x$. It is not the digits 3 and x written side by side.
- You can only collect like terms: $3x + 2x = 5x$, but $3x + 2$ stays as $3x + 2$.
- x on its own means $1x$, so $x + x = 2x$ — not x^2 .
- Multiplying is different from adding: $x \times x = x^2$, but $x + x = 2x$.

Worked example

Simplify: $2x + 3 + x + 1$

Step 1 — Like terms: $2x$ and x are like; 3 and 1 are like.

Step 2 — Group them: $(2x + x) + (3 + 1)$

Step 3 — Combine the x terms: $2x + x = 3x$

Step 4 — Combine the numbers: $3 + 1 = 4$

Answer: $3x + 4$ (check $x = 5$: $2 \times 5 + 3 + 5 + 1 = 19$ and $3 \times 5 + 4 = 19$)

Visual aid — collecting like terms

x	x	1	1	1		$2x + 3$
x		1				$x + 1$
x	x	x	1	1	1	$= 3x + 4$

Tiles of the same kind join together: $2x + 3$ and $x + 1$ give $3x + 4$.

Questions

Simplify each expression — show your working in your workbook

Q1 of 8

[1] ● ○ ○ ○ ○ ○

Simplify: $a + a + a + a$

Q2 of 8

[1] ● ○ ○ ○ ○ ○

Simplify: $5x + 2x$

Q3 of 8

[2] ● ● ○ ○ ○ ○

Simplify: $5x + 2x - 3x$

Q4 of 8

[2] ● ● ○ ○ ○ ○

Simplify: $7y + 4 + 2y + 5$

Q5 of 8

[2] ● ● ● ○ ○ ○

Simplify: $8m + 3n - 2m + 5n$

Q6 of 8

[2] ● ● ● ○ ○ ○

A pencil costs p pence. Write an expression for the cost of 4 pencils and a $30p$ rubber.

Q7 of 8

[3] ● ● ● ● ○ ○

Expand and simplify: $3(x + 2) + 2x$

Q8 of 8

[3] ● ● ● ● ● ○

A rectangle has length $(x + 3)$ and width x . Write a simplified expression for its perimeter.